Lunar Relics 3D — The Hardware We Left Behind

NASA Space Apps Cairo 2026 — Team Proposal
Selected challenge: *Abandoned but not Forgotten: Storytelling about NASA's Discarded Equipment on the Moon and Mars* 1
Live prototype: https://lunar-relics-3d.vercel.app
Repository: https://github.com/yousofahmad/lunar-relics-3d

Executive Summary

NASA's missions left landers, rovers, instruments, descent stages, and spacecraft on the Moon, Mars, and elsewhere. The selected challenge asks teams to turn those artifacts and the science they enabled into an engaging story for school-age and casual space enthusiasts 1. The information exists, but it is often distributed across mission records, technical pages, photographs, and 3D asset libraries. A learner may find a fact without seeing the machine, its parts, or its place in the mission.

Lunar Relics 3D is a bilingual Arabic-English web experience that makes the hardware explorable. The current prototype contains 16 structured mission records from NASA and NSSDCA sources. Visitors can browse Moon, Mars, and deep-space views; open a mission story; inspect a 3D representation; read a timeline; explore hardware hotspots; listen to optional Arabic or English narration; answer a short quiz; and earn a locally saved badge. The interface labels a NASA-published mesh separately from a reconstruction based on NASA photographs and specifications, so visual interpretation is not presented as an official model.

We chose this challenge because it combines historical storytelling, engineering, and planetary science in a form that can meet learners on a phone. The repository already provides the interactive foundation. During the hackathon, we plan to audit the 16 records against their linked sources, improve the story and accessibility review, test the experience on target devices, and document the provenance and licensing of assets. This approach meets the challenge by connecting each artifact to what it was built to do, where it operated, and what people learned from it.

Problem Definition

The challenge is not a lack of space-history data; it is a lack of an accessible connection between data, hardware, and science. Mission records are spread across technical archives and pages written for specialist readers. A learner may know a rover's name but not recognize its instruments, understand its mission journey, or see why its measurements mattered. The

challenge specifically calls for accurate, story-driven education about equipment left on the Moon, Mars, and elsewhere 1.

Lunar Relics 3D addresses this access problem with an object-centered interface: each record combines a sourced story, mission timeline, hardware explanations, bilingual text, narration, a quiz, and a 3D representation. It does not claim to replace NASA's primary records; it provides a readable route into them.

Background & Literature Review

NASA's 3D Resources hub provides free mission-related models and mirrors them on GitHub 2. NASA and NSSDCA mission records provide the dates, destinations, spacecraft identities, and source material behind the prototype's 16 records. These resources are authoritative, but they are not presented as one bilingual, object-centered learning path. The official challenge identifies the same opportunity: translate complex historical and scientific data into engaging narratives for younger and casual audiences 1.

The scientific record also shows why a story should connect equipment to evidence. Arvidson et al. describe how Spirit's position and physical properties were derived from wheel turns, suspension and tilt measurements, and overlapping images; they also report dust effects on solar output and observations of basaltic materials and water-related processes 3. A useful educational interface therefore needs more than a spacecraft picture: it should explain what a component did and what observations made the science possible.

Lunar Relics 3D fills this presentation gap rather than claiming a new scientific dataset. Its planned review will assess whether each story remains faithful to its linked primary sources.

Methodology

Current foundation. The repository stores 16 relic records in JSON. Each record includes bilingual story text, mission dates and status, location data where available, NASA/NSSDCA source links, hardware hotspots, a timeline, questions, a quiz, badge text, and Arabic/English narration paths. React and Vite provide the application structure; Three.js and react-three-fiber render the interactive scenes. An optional API can update records, while the local archive remains the fallback.

Planned hackathon process.

1. Audit each record against its linked NASA/NSSDCA sources and mark any uncertainty instead of filling gaps with assumptions.
2. Edit the narratives for age-appropriate clarity while retaining the distinction between documented fact and first-person storytelling.

3. Review Arabic and English content together, including right-to-left layout, subtitles, labels, and narration controls.
4. Check model provenance: keep NASA-published meshes distinct from SI-generated models built to NASA's published specifications.
5. Test the deployed web experience on desktop and mobile sizes, including loading, keyboard navigation, reduced motion, audio controls, and quiz/badge behavior.
6. Publish a short source and asset note with the final project and record what remains future work.

Solution

Lunar Relics 3D presents a searchable archive of 16 mission records across the Moon, Mars, and deep space. A visitor can select a mission from the map or archive, read its bilingual first-person story, follow its mission timeline, inspect labeled hardware hotspots, and consult NASA source links. Optional narration is available in Arabic and English. A short question turns each visit into an active learning step, and a badge records progress locally in the browser.

The current implementation uses unmodified NASA-published GLB assets for 11 missions; the 5 missions without a public NASA mesh (Apollo 15 rover, ALSEP, Surveyor 3, Sojourner, Phoenix) use SI-generated 3D models built to NASA's published specifications, each labeled on screen as "SI-generated · not from NASA". Some records reuse a representative design model, such as Opportunity for Spirit; the interface and source notes identify that limitation. This is a presentation tool, not a scientific simulator: planetary imagery, lighting, camera framing, and representative coordinates are used for exploration and are not presented as precision measurements.

Planned implementation work: complete the source audit; refine the content and bilingual review; improve low-end-phone performance and accessibility checks; and prepare concise documentation for sources, model provenance, and asset use. A classroom mode or printable cards may be explored only after the core submission is stable.

Value Proposition

For a learner, teacher, or family using a phone, Lunar Relics 3D offers one readable path from an artifact's name to its purpose, journey, hardware, and science. It is more approachable than navigating separate technical pages, more spatial than a text-only archive, and more transparent than an undifferentiated 3D gallery because model provenance is shown in the experience.

The bilingual interface and narration lower a language barrier for a Cairo audience without requiring an app install or account. The structured record format also makes the project

extensible: adding a mission means adding a documented record and its assets, not rebuilding the application. The product's value is educational access and honest presentation of NASA-related sources, not a claim that the prototype replaces those sources.

Role of Team Members

Exact names and final role assignments were not supplied. Use the following responsibility areas only after matching them to the actual team roster; do not infer or add members.

Responsibility area	Contribution
[TEAM MEMBER 1 — NAME TO BE PROVIDED]	Product scope, challenge fit, and final review.
[TEAM MEMBER 2 — NAME TO BE PROVIDED]	Mission research, source review, and bilingual content.
[TEAM MEMBER 3 — NAME TO BE PROVIDED]	3D asset sourcing or reconstruction and visual design.
[TEAM MEMBER 4 — NAME TO BE PROVIDED]	React/Vite front-end and Three.js interaction.
[TEAM MEMBER 5 — NAME TO BE PROVIDED]	Narration, subtitles, Arabic/English presentation, and accessibility review.
[TEAM MEMBER 6 — NAME TO BE PROVIDED]	QA, deployment checks, documentation, and submission materials.

If fewer or more people take part, merge or split these responsibility areas rather than creating placeholder members beyond the official roster.

Workflow Strategy

We will work in short vertical slices: select a mission, verify its source record, refine its story and labels, test the corresponding interaction, and review it before expanding the scope. The repository keeps content and interface data together, allowing a source correction and its presentation change to be reviewed as one unit.

Checkpoint 1 — analysis: confirm the selected challenge, freeze the record schema, and identify claims or assets requiring review.
Checkpoint 2 — build: improve the archive, mission story, hardware inspection, bilingual

controls, and quiz flow.
Checkpoint 3 — quality: test representative Moon, Mars, and deep-space records on desktop and mobile; review audio, keyboard access, reduced motion, and model labels.
Checkpoint 4 — submission: deploy the stable build, verify public links, finish the proposal and award disclosure, and record the final 90-second video according to the local guideline.

Communication, ownership, and exact staffing are **to be confirmed when the team names and roles are supplied**. The current prototype is the starting point; the checkpoints above are planned hackathon work, not claims that every item is already complete.

Resources

Software currently represented in the repository: React 18, Vite, Three.js, react-three-fiber, drei, the Web Audio API, npm scripts, GitHub, and Vercel hosting. The repository includes a production build command and automated Node tests for selected experience and asset-integrity behaviors.

Hardware: [TO BE CONFIRMED BY THE TEAM]. No microphone, phone inventory, laptop count, or other physical equipment is claimed here until the team provides those details.

Data and media: NASA and NSSDCA mission records and source links stored with the relic data; NASA's 3D Resources hub and the repository's model provenance file; local GLB assets; local Arabic and English narration files; and local Moon and Mars visual assets. NASA attribution and the distinction between published meshes and reconstructions must remain visible in the final release.

References

Team

Team name: [TEAM NAME TO BE PROVIDED]

#	Member name	Role
1	[MEMBER NAME TO BE PROVIDED]	[ROLE TO BE CONFIRMED]
2	[MEMBER NAME TO BE PROVIDED]	[ROLE TO BE CONFIRMED]
3	[MEMBER NAME TO BE PROVIDED]	[ROLE TO BE CONFIRMED]
4	[MEMBER NAME TO BE PROVIDED]	[ROLE TO BE CONFIRMED]
5	[MEMBER NAME TO BE PROVIDED]	[ROLE TO BE CONFIRMED]
6	[MEMBER NAME TO BE PROVIDED]	[ROLE TO BE CONFIRMED]